Fiscal Year:	FY 2005	**Task Last Updated:**	FY 11/01/2005
PI Name:	Charles, Harry K. Ph.D.		
Project Title:	Ground Based Measurement of Bone Loss in Astronauts Using AMPDXA GCS		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Technology Development Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
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PI Email:	harry.charles@jhuapl.edu	**Fax:**	FY 240-228-6119
PI Organization Type:	UNIVERSITY	**Phone:**	240-228-8050
Organization Name:	The Johns Hopkins University Applied Physics Laboratory		
PI Address 1:	11100 Johns Hopkins Road		
PI Address 2:			
PI Web Page:			
City:	Laurel	**State:**	MD
Zip Code:	20723-6099	**Congressional District:**	3
Comments:			
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No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date is 9/30/2007, per K. Major/NSBRI (12/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Feldmesser, Howard (The Johns Hopkins University) Beck, Thomas (The Johns Hopkins University School of Medicine) Magee, Thomas (The Johns Hopkins University) Chen, Michelle (The Johns Hopkins University Applied Physics Laboratory) Spisz, Thomas (The Johns Hopkins University Applied Physics Laboratory)		
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Task Description:	The Johns Hopkins University Applied Physics Laboratory and the School of Medicine are building a Ground-based Clinical System using Advanced Multiple Projection Dual-energy X-ray Absorptiometry (AMPDXA) principles to go beyond bone densitometry to measure the structural mechanics of bones and muscles. Exposure to microgravity during long-duration space flight decreases bone and muscle mass but the implications for injury differ from similar losses on earth and cannot be discerned by densitometry. The goal of the Advanced Multiple Projection Dual Energy X-ray Absorptiometry Ground-Based Clinical System (AMPDXA-GCS) is to produce an operational instrument that can be transferred to the NASA-Johnson Spaceflight center for use in the pre-and post-flight bone mineral density (BMD) and bone structure measurements on astronauts. The original aims of the project include: 1. Design, fabrication, and testing of AMPDXA-GCS. 2. Validate the utility, radiation safety, and method of operation of the AMPDXA-GCS configuration. 3. Address key technical challenges necessary to migrate the AMPDXA-GCS from ground-based studies to spaceflight. 4. Transfer AMPDXA-GCS technology into commercial clinical uses. The AMPDXA-GCS will be able to detect 1% changes in bone mass and geometry and 2% changes in muscle mass and it will permit biomechanical assessments with similar accuracy. It will also decompose soft tissue into fat and muscle. Its accuracy and broad measurement capability make the AMPDXA-GCS superior to any conventional means for assessing bone health and strength. An imaging system with AMPDXA-GCS qualities in space would be an unrivaled tool for rapid, high-precision testing of the effects of countermeasures on the musculoskeletal system. Thus, AMPDXA-GCS technology is highly likely to result in new tests for countermeasures efficacy and it promises to make a significant contribution to NASA Critical Path Roadmap (CPR) risk reduction. AMPDXA-GCS monitoring techniques could also benefit millions of people suffering from osteoporosis and other bone disorders. Research and clinical medicine will both benefit from increased ability to more precisely monitor bone health and the efficacy of treatments. Plus, the design principles will allow the development of a compact less expensive version for limb imaging that could be conveniently used in a variety of remote settings such as rural clinics and nursing homes. To achieve the eventual space flight version of the AMPDXA-GCS, it will be necessary during the AMPDXA-GCS development to address remaining technical challenges associated with building a low power space qualified system weighing 45 to 50 kilograms (kg). Key among these challenges are development of a high-speed, lightweight, kilo-voltage (kV) switching power supply, an advanced high-efficiency x-ray source, and a compact and lightweight scanner configuration. The proposed aims will be achieved by leveraging off prior successful NSBRI funded collaborations between the Johns Hopkins University Applied Physics Laboratory (APL) and the School of Medicine (SOM) which resulted in an AMPDXA Laboratory Test Bed (LTB) and a Human Test Bed (HTB). The LTB has been fully operational for the last six years. It was used to develop sources, detectors, and software algorithms necessary for high-precision measurement of BMD and bone structure. Most recently, it was used to refine BMD and structure extraction algorithms and for continued progress on high-resolution separation of soft tissue from bone. It has been modified to mimic the operation of the AMPDXA-GCS to verify the new detection geometery. Initial images have been taken and the results are consistent with projections, verifying the new 3 simultaneous beam mode of operation. The HTB has been operational for three years with several human imaging trials conducted. The HTB delivered multiple projections, high-resolution DXA images. Results using phantoms and cadaver parts have shown precision comparable with the highly accurate LTB. The design used in the HTB permitted multiple projections about the fixed rotational axis of the scanner, but experiments showed that greater flexibility in projection geometry is needed. The AMPDXA-GCS builds on HTB with better 3-D geometry capability and it is also more consistent with full body commercial DXA scanners in basic configuration and cost. Key plans for the coming year include: 1) operation of the AMPDXA-GCS testbed, 2) design of the AMPDXA-GCS, 3) procurement of remaining parts, 4) structural fabrication of AMPDXA-GCS, and 5) the start of electronics integration.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The AMPDXA project has many implications for future research and development. The AMPDXA-GCS, as described above, has direct application to risk reduction in NASA's Critical Research Path. The AMPDXA-GCS is capable of real-time monitoring of bone and muscle loss, at any anatomical location, with extremely high precision. Since the results are patient-specific and not tied to volumetric averages and statistical norms, the AMPDXA-GCS is a useful tool for monitoring the effectiveness of countermeasures as well as determining the risk of fracture under various loading conditions and activity scenarios. The AMPDXA-GCS also appears to be a natural adjunct to earthbound research on the effect of aging and disuse on bone integrity. It could also be used as a routine screening tool for osteoporosis and as a monitoring instrument for osteoporosis drug therapy. Since the AMPDXA-GCS measures structure to high precision and does an excellent job at imaging not only bone but also manmade prosthetic implants, the AMPDXA-GCS could be used by orthopedic surgeons to study the life progression of implants (e.g., loosening, bone regression, etc.). Given its low radiation dose and ability to automatically determine dimensions from the collected radiographic data, the AMPDXA-GCS may prove valuable in other orthopedic applications as well. For example, it is possible to predict stress fractures from BMD measurements. Stress fractures in military recruits range from one to seven percent, depending upon gender and branch of service. An untold number of athletes and other individuals also suffer stress fractures annually as a result of intense repetitive exercise. Such exercise or other similar activity can tire muscles so that they are no longer able to absorb the shock from these activities. As a result, tiny hairline cracks known as stress fractures develop in the bone. The best treatment for stress fractures is rest for six to eight weeks. Such rest periods would prevent a professional athlete or a soldier from performing his/her job. Thus, the ability to identify persons susceptible to stress fractures would enable a different training regimen for these persons in order to prevent the onset of a stress fracture. Commercially, if cost projections can be realized, there is a significant market for population screening and treatment monitoring of osteoporosis. Based on the AMPDXA-GCS, an easy-to-use, relatively compact instrument could be developed for a small clinic or doctor practice. The unit could be either horizontal like the GCS or vertical (patient stands). Concepts for a vertical unit have been developed. The market for such a commercial AMPDXA could range from several hundred to several thousands of units, depending upon final cost. If the unit could be structured to be lightweight and relatively transportable, it would have important implications for use as a tool in nursing homes, clinics, and doctor's offices. Several key members of the NSBRI's Bone Research Team have already indicated that a precision AMPDXA would greatly assist their work in bone loss studies, especially in bed rest studies. AMPDXA commercialization efforts are being pursued by both the APL Technology Transfer Office and the Technology Transfer Office at the Johns Hopkins Medical Institutions. Commercialization of the AMPDXA software for use with conventional DXA data may provide the stepping-stone for the commercialization of the AMPDXA hardware.

Task Progress:	A bench-top version of the AMPDXA-GCS system has been fabricated to test out the physical principals of the new AMPDXA system. The system is operational and images of cadaver bones and phantoms have been made. A new higher accurate phantom for new geometry has been deigned and is being fabricated. The x-ray tube produces a beam that is parallel to the plane of the table and detected by 3 detectors positioned at zero, minus 15, and plus 15 degrees apart. Lead collimators are used to limit the beam to just the area of the detectors. A linear motion stage is used to move a test sample through the beam which simulates the motion of the C-arm relative to the patient in the final AMPDXA-GCS design. The bench-top also allows adjustment of tube to sample (¡§patient¡¨) distance in order to optimize the final design. The tube to detector distance can also be varied on the bench-top as part of this optimization. A laboratory jack is also provided to vary the distance along the sample (patient) axis and allows multiple images to be integrated together in software to verify physical concepts before design and fabrication of the AMPDXA-GCS begins. Three sensor array electronic units have been assembled, tested and electrically evaluated. All pixels have been mapped per the agreed upon order and this mapping has been tested. Using the test mode at 1 ms sampling with the 50 pico Coulomb (pC) (50 nA) range, a 13 pC charge gives a reading of around 2800. These electronic units are each uniquely identified USB (Universal Serial Bus) devices. They are powered from a four port USB hub. Each sensor array unit draws 280 ma @ 5V quiescent, 300 ma running. It was decided to make the beam monitor a separate USB device. The microcomputer and analog board designs were merged onto one board and connectors for the scintillation detector were added. This board is expected to draw around 55 ma. Although the A/D used on the analog boards allows for direct input of current source devices, a transconductance amplifier option was added to help scale the detector¡¦s output, should this be necessary. The beam monitor board has been designed, manufactured and installed on the AMPDXA-GCS laboratory test bed and is operational. Currently, the data is digitized every 1 ms. Since signal integration takes 350 ƒÝs for all channels and data collection takes about 215 ƒÝs. this leaves only 435 ƒÝsec for moving data to the USB engine and handling data collection control. Going slower won¡¦t be a problem. Since each USB device is currently a stand-alone data collection device, it has its own unsynchronized 12 MHz clock. Observed relative stability of two of the clocks showed a 1ms/200s drift which is 5 PPM, better than the crystal specification. This means that within 200 s we could drift 1 sample time. The start of conversion times will also vary up to 1 ms currently as their phases
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